

PAPER_706: NGC 3603 Star Cluster (Primary Analysis): Gas Conversion UQFF MUGE

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Abstract

Primary analysis of the NGC 3603 young super star cluster with emphasis on gas-to-star conversion efficiency and Hubble expansion correction. The cluster contains WN6h Wolf-Rayet stars and has triggered secondary star formation in surrounding Bok globules. The primary UQFF MUGE is parameterized by effective mass evolution with gas consumption.

1. System Parameters

Parameter	Value	Description
M_{base}	$4 \times 10^5 M_{\odot}$	Base cluster mass
r_{clust}	$9.0 \times 10^{15} \text{ m}$	Cluster radius
f_{gas}	0.30	Initial gas fraction
α_{wind}	0.05	Wind suppression
B_{clust}	$1.2 \times 10^{-5} \text{ T}$	Cluster magnetic field

2. Master UQFF Gravity Equation

$$g_p(t) = \frac{GM_{eff}(t)}{r^2} (1 + H_z t) (1 - \alpha_{wind}) (1 + f_{TRZ}) + q(v \times B) \left(1 + \frac{\rho_{UA}}{\rho_{SCm}} \right) \times 10^{-12}$$

2.1 Effective Mass with Gas Conversion

$$M_{eff}(t) = M_{base} (1 + f_{gas} e^{-t/\tau_{SF}})$$

3. References

- Brandl et al. (1999), A&A, 352, L69
- UQFF Framework, Star-Magic Session 175

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